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FMCAD 2026 Student Forum: Notification for paper 228

1 message

FMCAD 2026 Student Forum <fmcad2026studentforum@easychair.org> 22 July 2026 at 06:43
To: Sudrit Ghimire <sudritghimire@gmail.com>

Dear Sudrit Ghimire,

On behalf of the FMCAD 2026 Student Forum Program Committee, we are happy to inform you that the following submission has been accepted for presentation:

TimeFence: Practical Sequence Evidence for Linux Telemetry

Congratulations! All papers underwent a thorough peer review process with three reviews each. These reviews and Program Committee discussions were carefully considered in the final decisions.

Please find the full text of the reviews for your submission below. We hope that the provided feedback will help improve your work and prepare the presentation for the Student Forum.

We will be in touch shortly with more details regarding the format of the presentation and poster.

Please note that all student forum participants are required to register for the main FMCAD conference. Early registration with reduced pricing ends on August 1.
Please reach out to us if you have any questions.

Thank you for submitting to the FMCAD 2026 Student Forum. We look forward to seeing you at TU Graz!

Best Regards,
Tanja Schindler and Lee Barnett
FMCAD 2026 Student Forum Co-Chairs

SUBMISSION: 228
TITLE: TimeFence: Practical Sequence Evidence for Linux Telemetry

---------- REVIEW 1 ----------

SUBMISSION: 228
TITLE: TimeFence: Practical Sequence Evidence for Linux Telemetry

---------- Overall evaluation ----------
SCORE: 1 (weak accept)
----- TEXT:
Summary
The paper introduces TimeFence, which uses “K-Sentry”, a verified finite-field accumulator, to compute order-sensitive digests of telemetry data. The technique is used for determining whether telemetry data (after it leaves the collection path) has been manipulated in any way (e.g. events being dropped, processed out-of-order, new events being inserted etc.), which can impact the use of the data for incident response and auditing purposes. The arithmetic invariants of the accumulator are verified in Verus. The overhead of the system is evaluated on 5 different workloads.

Evaluation
The problem is clearly stated, is well-motivated, and distinguished from existing tools, such as Falco and auditd. The verification boundary section is great in that it describes the scope of what Verus covers and doesn’t cover along with the limitations of K-Sentry. The preliminary evaluation shows promising results of TimeFence when compared to Falco and auditd.

The bulk of the paper focuses on the “systems” aspects and less so on the formal aspects beyond stating that the accumulator core is verified in Verus. For example, I would have liked to see more details on soundness against tampering of events. I’d encourage the authors to focus more on the “verified” aspects of the approach to make the FMCAD fit more clear.

Comments:
* FMCAD student forum allows trimming the related works section to what's crucial, but the paper does not include any. It would be nice with a least a couple of references to get more context on the contribution
* There are no references in the paper, e.g. auditd, Falco, and Verus. Can you add citations?
* In table 1: What does "Checkpoint localization" and "19/19 at interval 1000" mean?
* The abstract mentions that the range of the overhead is 0.83 - 6.67% but in table 2, the "exec" workload has -0.31%. Should one of them be corrected? What's the intuition behind a negative overhead?

---------- REVIEW 2 ----------

SUBMISSION: 228
TITLE: TimeFence: Practical Sequence Evidence for Linux Telemetry

---------- Overall evaluation ----------
SCORE: 0 (borderline paper)
----- TEXT:
Summary

The paper is about TimeFence, a layer for Linux telemetry. The goal is to get a streaming, compact, order-sensitive digest that can be checked later against evidence certificates. To this end, a verified finite field accumulator is employed. Several experiments are presented.

Evaluation

Overall, the structure of the paper is clear and the work seems relevant, but the presentation could be improved in the following points.

* The extended abstract is very hard to read because it is basically assumed that the reader knows the approach. Core terminology needs to be introduced.

* There are a lot of formulas, which are not well explained. All the symbols that occur in the formulas need to be introduced.

* The status of the approach is not clear. Is there already some tool/paper published?

* Table 1 does not seem to be referenced in the text - some integration would be beneficial.

* It is hard to assess what we can learn from the experiments.

* No references are given. Some other tools are mentioned, but the state of the art that is improved by this research should be made more explicit.

* Finally, the overall research question addressed in this work need some more elaboration.

---------- REVIEW 3 ----------

SUBMISSION: 228
TITLE: TimeFence: Practical Sequence Evidence for Linux Telemetry

---------- Overall evaluation ----------
SCORE: 2 (accept)
----- TEXT:
The paper presents an approach for computing a hash-like summary of a log of events that is sensitive to various classes of modifications, including changing the order of events. This approach is evaluated in the context of Linux kernel telemetry.

Overall, the paper presents an interesting approach with solid experimental results. The implementation is also verified in Verus.

---------- FEEDBACK ----------
There is no author feedback for this paper.